Science

Nature Walks & Scouting
General Science
Labs
Natural History
Nature Notebook

Alveary

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Science: Grade 8

GRADE(S): 8

About the Course

In level 8, learners study science in its historical, political, and cultural context with a growing understanding that science is an ever-changing process rather than a static body of knowledge. Reading level and expectations increase from previous levels.

This course includes the following topic(s): General Science: Grade 8, Natural History: Grade 8, Labs: Grade 8, Nature Notebook: Grade 8, Nature Walks & Scouting: Grades 1-8

About Nature Walks & Scouting: Grades 1-8
Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

About General Science: Grade 8
This is the second of a two-year Physical Science course (i.e., introductory physics and chemistry). The fullness of the course includes topics in earth science, history, politics, and more. The completion of this course provides a gentle yet robust transition into the high school disciplines. The lab component is required for this course. Coordinating afternoon activities are provided in Outdoor Work.

About Labs: Grade 8
This is the required lab component for level 8 General Science. This course includes our custom laboratory book which will build foundational and progressive skills and habits.

About Natural History: Grade 8
Learners explore considerations from the history of microbiology and infectious disease. Coordinating afternoon activities are provided in Outdoor Work.

About Nature Notebook: Grade 8
Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

Science: Grade 8
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. In Form 1, students became acquainted with this area of knowledge, science. Form 2 students extended their scope and began to encounter science as an active part of society. Form 3 students extend their relationship with science in a new direction (time) and consider science as a process by which man seeks Truth within Creation. They begin to situate science in its historical, political, and cultural context with all of the complexities of man's own story. In this way, students are prepared to think about science with a complete and holistic perspective.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

Science: Grade 8
Learners may stay at their appropriate level for General Science while combining Natural

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History and Outdoor Work, or teachers can adapt the laboratory content and reading level to meet their needs.

Nature Walks & Scouting: Grades 1-8
Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

General Science: Grade 8
For eighth-grade students or possibly hungry seventh graders or ninth graders taking their time. If combining grades is desirable: the combined grades may begin at the appropriate level and move forward together, or teachers can adapt the reading level and laboratory content to meet their needs.

Labs: Grade 8
The ideas and skills in this component are progressive, like math or grammar. Teachers should read the lab book thoroughly to understand what concepts might need to be supplemented should they choose a different sequence or a substitution.

Natural History: Grade 8
For approximately eighth-grade students based on a balance of complexity and reading level. However, learners may be freely combined based on needs and preferences.

Nature Notebook: Grade 8
Learners may be combined and share prompts or follow their own interests. Support through varied media, scribing observations, or notebooking in tandem.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
1-8	Nature Walks & Scouting: Grades 1-8 1 time/week 30 min+	
8	General Science: Grade 8 3 times/week 30 min	The Story of Science: Einstein Adds a New Dimension The Story of Science: Newton At the Center
8	Labs: Grade 8 1 time/week 45 min	Science: Grade 8 Lab Book
8	Natural History: Grade 8 1 time/week 30 min	Invincible Microbe
8	Nature Notebook: Grade 8 1+ time/week 20 min+	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 8				
Natural History: Grade 8	General Science: Grade 8 Nature Walks & Scouting: Grades 1-8	General Science: Grade 8	General Science: Grade 8 Nature Notebook: Grade 8	Labs: Grade 8

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 8
☐ Obtain any supplies indicated on the science or grade-level supply lists.

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☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings, except where the suggestion is to collect some Thing for a lesson.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Labs: Grade 8
☐ Preview science labs and purchase materials or have students gather them. Print or shortcut any links, as desired.

Special Topics & Field Trips

Science: Grade 8
Encourage students to choose their own special topic and to notice its ecological relationships. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too!

☐ Learn a few of your local species or varieties in connection with special topics.

General Science: Grade 8
☐ Term 1: A science museum with motion exhibits
☐ Term 2: A science museum with chemistry exhibits
☐ Term 3: A planetarium or local astronomy club event

Natural History: Grade 8
☐ Any term: a science museum with an exhibit on microbes

Term Prep & Teacher Tips

General Science: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- 5 Starburst candies or similarly-shaped manufactured item
- sturdy wooden board at least 2-3" wide and 3.5' long to construct marble ramp
- anything to create rails for the ramp, more scrap wood, old pool noodles, etc.
- whatever is needed to attach the rails: bar clamps, hammer and nails, glue, etc.
- books or wood blocks to adjust height
- 3 smaller blocks to act as stops
- 1-6 eggs (Term 1)
- water
- large sheet pan or flat location outside
- 2 plastic cups, 1 smaller and 1 larger
- various objects to test for sliding on ramp, such as small objects made of wood, plastic, metal, etc
- various Lego-type blocks that can be used to build a small box, roughly 1-2" dimensions
- a larger bouncy ball, such as a basketball
- a location to bounce balls
- 1/2 c small solid fill material, such as cat food or rice
- 2 locations at 2 different temperatures, such as a sunny window and a chilly basement
- plastic wrap
- wire cutters, such as those on many pliers
- hammer
- at least 2 different light sources, including bright sunlight and at least 1 type of artificial light
- microwave
- microwave-safe casserole dish, at least 9"
- marshmallows to cover the bottom of the dish, Peeps or jumbo work best
- stopwatch, such as found on most electronic devices
- calculator, such as found on most electronic devices
- printables
- computer
- optional: hot glue gun
- optional: long forceps
- optional: matches
- optional: glass jar that is taller than a tea light candle
- optional: hard boiled egg (Term 2)

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Natural History: Grade 8
☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
- 8 Tbsp cornstarch
- 1-2 c milk (Term 1)
- 3-5 c white vinegar
- paper towels
- water
- 1 egg (Term 1)
- 3 antimicrobial agents used at home (e.g., soap, hydrogen peroxide, rubbing alcohol, essential oils, etc.)

Reminders

General Science: Grade 8
☐ Note that Lab in week 6 of Term 1 calls for fresh eggs and Lab in week 2 of Term 2 includes an optional activity that uses a hard-boiled egg. Make a note on your calendar if you will need to purchase these closer to the appropriate dates. Refer to the lab book for more details.

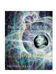
Natural History: Grade 8
☐ Note that the first lesson of Natural History is an activity, so be prepared with these materials and any others from the supply lists on day 1.
☐ Any type of milk is needed for Week 5. Make a note on your calendar if you will need to purchase this closer to the appropriate time.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.
∞ View Book List Details

Science: Grade 8

General Science: Grade 8

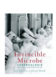

The Story of Science: Einstein Adds a New Dimension


The Story of Science: Newton At the Center

Labs: Grade 8

Science: Grade 8 Lab Book

Natural History: Grade 8


Invincible Microbe

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.
∞ View Basic Supplies
∞ View Supply List Details

Science: Grade 8

Alveary Grade 8 Science Kit

Nature Walks & Scouting: Grades 1-8


Wild Bird Seed


Bird Feeder


Hand Lens


Small Collection Containers


Small Basin

Dip Net

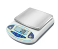

Student Microscope

General Science: Grade 8

Household Items - General Science: Grade 8

Labs: Grade 8


Small Collection Containers


Scale/Balance

Natural History: Grade 8


Prepared Microscope Slides

Household Items - Natural History: Grade 8


Slides and Coverslips


Student Microscope

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Quick Links

Click THIS text or scan the QR code for links.

Science: Grade 8

- Extra Helpings
- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android
- Alternate One Year Physical Science Lesson Plans
- Foundations (See Section 13: Science)

General Science: Grade 8

- AmScope key for slide IDs

Labs: Grade 8

- Grade 8 Lab Book
- Lab Notebook Examples

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Science: Grade 8

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

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Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

- These can be viewed as alternative ways to engage.

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Science: Grade 8

How To Teach Labs

Introduce

Regardless of how many days are required to complete a particular activity, every science lab has the same flow, which follows the scientific method and is guided by the lab book.

• On day 1, learners read an introduction in their lab book. How does this lab activity relate to what they have learned so far, as well as any previous experience? What will they learn from the lab? This is analogous to the conversation we might have when we begin something new in any subject, but they may need to dialogue as they learn to extend these skills to the laboratory.

• Once they have had a chance to think, they will compose a prelab narration to put these introductory ideas into their notebooks. The prompt in the lab is generalized and consistent, so they learn the habit. Eventually, they will learn to formulate this as a hypothesis. For example, a learner preparing for a lab about the use of insect repellent might write:

"I have read about some diseases that are spread by insects, like Lyme disease. I also know that my sister is allergic to some insect bites. Insect repellent contains pesticides to keep insects away. Some scientists worry about how pesticides affect wildlife. I am going to compare some different insect repellants in this lab to see if they really work."

• Written narration and composition are skills that they will build over time. These prelab narrations may seem short and even incomplete at first, but that is okay. If learners have difficulty or are easily frustrated, then provide them with support. Teachers may act as scribes or allow students to keep a digital notebook to type or use assistive technology, as appropriate.

• After they complete their prelab narration, the listed materials are collected. This gives the learner some responsibility to let the teacher know if something is missing or to remind the teacher if something needs to be purchased at the store.

• If these activities on day 1 do not fill the scheduled time, that is fine. They might use the additional time to familiarize themselves with the procedure, draw a picture from their book, or catch up on any other work. Some labs may instruct the student to begin on the same day.

Lab Procedure

• Then, students begin and follow the procedure (whenever prompted by the lab book). Note that the lesson plans guide teachers as to which sections are completed each week, and the lab book instructs learners when to take a break.

• The lab gives instructions for using their notebooks to create tables and figures, as needed. Do not allow this to become an obstacle. Do it with them, first having them watch and then having them copy or help when ready.

• If teachers choose to have learners record directly in the lab book or on a photocopy, then cut out and tape these into their lab books, so that they can see how the record is built.

• Learners may feel unsatisfied with their results. This is often part of the process. Come alongside, help them learn to be okay with uncertainty and questions, and teach them what to do with it in the next step.

Analysis & Conclusions

• The last step in the lab is to analyze the data and observations and draw conclusions from them. For some simpler labs, learners will complete their analysis and concluding (or postlab) narration on the last day of the lab procedure. For labs that are more involved, a separate day has been built in to allow adequate time for this.

• Similar to the prelab narration, the concluding narration is a chance to think about and put into

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words (or questions). This time, they are considering what they learned from the lab, what they could learn more about if they were to continue, and possibly how they would pursue that learning. Again, they might need support in the form of dialogue, a scribe, etc. For example, the above learner might write:

“It was clear that the Off and black pepper essential oil worked against ants because they would not even touch the line of repellent, but I wasn’t sure about the Skin So Soft because it ran all over the place. I could test this one again with different insects or on a different surface.”

● Depending on the interest of the learner and the priorities of the teacher, the student might be encouraged to spend more time on those ideas of what more they could learn, or it might be time to move on. Either way, it is an important part of the scientific method to reflect on what we could or would do next - our practice should help to clarify our thinking and teach that there is always more to be learned.

● Teachers should engage learners with this reflection by reviewing their lab notebooks with them, discussing the science used in the lab, and demonstrating curiosity about the lab themselves.

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Science: Grade 8

[Click THIS text or scan the QR code for links.](#)

Term 1

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 1**
Cell Activity

▯ Materials: toothpick, glass slide and cover slip, candle, methylene blue stain, water, paper towel, microscope, egg, cup/glass, white vinegar

→ INTRO
Most of your experience in natural history has been with Things that you can see around you. The invention of the microscope led to the discovery of a whole new world. This world of microscopic life is our focus this year. We begin by better understanding the microscopic cells from which you are made. First, you will make a slide of your cheek cells. Then you will set up an egg as a simple model of the cell.

→ ACTIVITY
• Using a toothpick, gently rub the inside of your cheek.

• Now rub the toothpick on the surface of a glass slide, keeping the smear near the middle of the slide. Let this dry for a moment on a paper towel while you light your candle.

• Heat fix your slide by passing the slide through the flame 3-4 times. The underside of the slide should pass through the flame, but never stop. This will help your sample stick to the slide better. When you are finished, you can blow out the candle.

• Using protection, squeeze a drop of methylene blue stain onto your cheek smear. This stain will stain everything, so you MUST protect your clothing and anything else you do not want to be stained! Let the slide with the drop of stain sit for about 2 minutes.

• Gently rinse from the back of the slide with a little clean water and carefully blot dry. Do not rub your smear.

→ OBSERVE & NARRATE
• Set a cover slip on your slide and examine it under the microscope, slowly increasing the power as you want to see more. Draw or diagram what you see in your science notebook.

→ ACTIVITY
• Now, place an egg inside a cup or glass and cover with vinegar. Replace the vinegar every day for three days. Once the shell has been dissolved, the egg can be left on a paper towel until your next lesson.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 1**
Setting the Stage

▯ Materials: Newton at the Center

PREP: Read Teacher Tip

→ INTRO
This book tells the story of how physical science developed from the Renaissance into the 1800s. As you move through the story, notice how ideas and knowledge about the world around us changed through a combination of imagination and experimentation. There are many interesting sidebars and tangents in the text, but read the main text first and then explore them, as desired. Watch the course introduction video:
∞ Video Link: Form 3 Science Intro (5:04)

→ READ, NARRATE, & DISCUSS

★ TEACHER TIP
Watch the course introduction video with your student(s).

★ TEACHER TIP
As we will begin some notebooking next week, help students notice the author's structure and purpose this week during discussion.

• IMPORTANT DATES
Beginning of the Renaissance (c.1453)

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Science: Grade 8

[Click THIS text or scan the QR code for links.]

Term 1

Ch.1 p.1-6 "For centuries" - "good place to start."

• What do you gather so far about the author's purpose for this chapter?

WEEK 1

30m General Science: Grade 8 - Lesson 2
Setting the Stage

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.1 p.6-13 "Imagine yourself" - "familiar landmarks."

• This chapter sets the scene for Copernicus. Describe that scene.
• Consider the relationship between church and state in Copernicus' time. Was a singular church and state good for either the church or the person? If so, in what ways? In what ways was it bad for both the church and the person? Is a separate church and state bad for either the church or the person? If so, in what ways? In what ways is it good for both?

★ TEACHER NOTE
A simplified contrast between medieval philosophical and theological beliefs and modern scientific beliefs is presented on the p.10 sidebar. Teachers might allow it to pass by the way, engage students in discussion, or skip this one (the larger ongoing picture takes shape over the year).

WEEK 1

30m General Science: Grade 8 - Lesson 3
Copernicus

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.2 p.14-20 "Copernicus" - "defines the age."

• Tell about Copernicus' life.
• Describe the shift in ideas observed by the author.

• IMPORTANT DATES
Nicolaus Copernicus (1473-1543)

WEEK 1

45m Labs: Grade 8 - Lesson 1
Measuring Starburst Candy

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of Measuring Starburst Candy, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and Procedure this week.

★ TEACHER TIP
Students measure manufactured candy to learn about metric, accuracy, and precision. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 2

30m Natural History: Grade 8 - Lesson 2
Cell Activity

Materials: egg from last week, Prepared Microscope Slide Set, microscope, image link

PREP: Read Teacher Tip

➔ INTRO
Today, you will compare your egg model to the cheek cells that you observed under the microscope, compare these to a variety of animal cells, and then to some bacterial cells.

★ TEACHER TIP
If needed, be sure to pour your agar plates for next week according to the package instructions.
∞ **Video Link:** Pouring Agar Plates (2:51)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 1

→ OBSERVE, NARRATE, & DISCUSS
• Notice that your egg model has an outer membrane and a well-formed internal structure (the yolk) that is suspended in a matrix (the egg white). Compare to your cheek cells from last week. How are they similar? How are they different?

• This same basic structure is used to accomplish a great number of tasks. View slides 14-20 from your Prepared Microscope Slide Set. Try to identify these basic parts in each slide. Look carefully at slide 15. Red blood cells lose their nucleus so that they can carry oxygen more efficiently. You may also be able to find some white blood cells on this slide. How do they look different?

• View the Cellular Diversity Slides at the link to see a variety of bacterial cells. What do you notice about the bacterial cells in comparison? Can you still identify the same basic parts? How do the cells compare in size? Draw or diagram your observations in your science notebook
∞ Link: Cellular Diversity Slides

WEEK 2

⬚ 30m General Science: Grade 8 - Lesson 4
Copernicus in the Renaissance

▯ Materials: Newton at the Center

PREP: Read Teacher Tip

→ INTRO
Recall what you read about Copernicus' life and the Renaissance setting for our story. As you continue reading, begin taking notes, jotting down the ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. If you'd like to see a few examples, watch the Notebooking video:
∞ Video Link: Notebooking? (1:39)

→ READ, NARRATE, & DISCUSS
Ch.2 p.20-31 "Since we have" - "and the earth."

• Describe the contradictions observed in Renaissance society.
• Discuss the scientific and religious ideas on Copernicus' mind.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. Invite them to share their notebook entries to improve the discussion!

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 2

⬚ 30m General Science: Grade 8 - Lesson 5
Leonardo da Vinci

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.32-35 "What's to be" - "and the Arno."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Leonardo da Vinci.

WEEK 2

⬚ 30m General Science: Grade 8 - Lesson 6
Copernicus Has a Problem

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

• **IMPORTANT DATES**
Copernicus published On the Revolutions of the Heavenly Spheres (1543).

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Science: Grade 8

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Term 1

→ READ, NARRATE, & DISCUSS
Ch.3 p.36-43 "Copernicus is" - "move the earth."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● What does the author mean by "thought experiment"?
● Discuss some of Copernicus' questions.
● Explain Copernicus's conflict over his ideas.

WEEK 2 ⬚ **45m Labs: Grade 8 - Lesson 2**
Measuring Starburst Candy

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 2 of Measuring Starburst Candy, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Analysis and Conclusions today.

★ TEACHER TIP
Inviting students to share their lab notebooks is a great way to engage discussion and keep up with their learning! They should be able to discuss the science and explain what they did.

WEEK 3 ⬚ **30m Natural History: Grade 8 - Lesson 3**
Microbe Activity

Materials: agar plates

→ INTRO
Because bacteria are so small, humans were unaware of their existence or their place in our world for a very long time. One way to see them more easily is to grow a colony of them. You will use petri dishes with a kind of food, called agar, to do this. The dishes will be left open at various locations for several hours to see what kinds of microorganisms are in the environment at each location.

→ ACTIVITY
● Choose three locations that you would like to test for the presence of bacteria. These might be your classroom, the bathroom, the kitchen, the playground, etc.

● Take four of your agar plates and, keeping them closed, turn them upside down on your work surface. Label the bottom of each plate using a permanent marker. One plate will be labeled "control," while the remaining three will be labeled with your three locations.

● Leave the control plate closed and where it is while taking the three experimental plates to their locations. For each experimental plate, open the lid and leave the plate right-side-up in that location. You will leave it there for about three hours.

● After several hours exposed, collect each plate, replacing the lid. You may want to secure all of the plates with tape. Then leave them in a warm (but not hot) place for the next week.

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 7**
Tycho Brahe Observes

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

● IMPORTANT DATES
Tycho Brahe (1546-1601)

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Science: Grade 8

[Click THIS text or scan the QR code for links.]

Term 1

→ **READ, NARRATE, & DISCUSS**
Ch.4 p.44-48 "The movement" - "is going on?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Tycho's early life.
- Explain why Tycho's approach was such an important development in thought.
- Why did people think that "perfect" meant "unchanging"? Discuss the implications of this idea.

WEEK 3

☐ **30m General Science: Grade 8 - Lesson 8**
Tycho Brahe Observes

Materials: Newton at the Center

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.4 p.48-57 "Only once" - "sending up sprouts."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about this episode in Tycho's life.
- Compare the models of Ptolemy, Copernicus, and Tycho using words, drawings, or diagrams.

★ TEACHER NOTE
The timescale of traveling light is mentioned on p.51: "It has taken 20,000 years..." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

● **IMPORTANT DATES**
Newton At the Center
Choose events from the timeline on p.56-57.

WEEK 3

☐ **30m General Science: Grade 8 - Lesson 9**
What is Parallax?

Materials: Newton at the Center

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.4 p.58-61 "When Tycho" - "a solar system."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain what parallax is and how it is used.

WEEK 3

☐ **45m Labs: Grade 8 - Lesson 3**
Galileo's Falling Balls

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ **LAB DAY**
Complete day 1 of Galileo's Falling Balls, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. As time permits, they may begin steps 1-3 of the Procedure.

★ TEACHER TIP
Students work with velocity and acceleration while practicing with metric and use of a spreadsheet.

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Science: Grade 8

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Term 1

WEEK 4

⬚ **30m Natural History: Grade 8 - Lesson 4**
Microbe Activity

▯ Materials: agar plates from last week

PREP: Read Teacher Tip

→ INTRO
Today, you will examine your plates! Which location do you think will have the most microbial growth?

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your four plates and examine them. Draw or diagram them in your science notebook. Hopefully, your control plate is still clean. Which of your experimental plates has the most growth? Describe the appearance of the growth. Are there dots of separate colonies? Is the entire plate covered? Does the growth all look the same, or are there different types of microbes, indicated by different colors and textures?

• It is important to understand that even though we are surrounded by microorganisms all the time, we usually don't get sick. Many microorganisms are harmless or even beneficial. Look around your home or school, or talk to adults to find three ways that microorganisms are beneficial. Examples are cheese, yogurt, any fermented foods, and certain health supplements.

★ TEACHER TIP
Next week's activity requires a brief prep that requires any type of milk.

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **30m General Science: Grade 8 - Lesson 10**
Enter Galileo

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.5 p.62-66 "Twenty-one years" - "in scientific circles."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Galileo's family and early life.
• Describe the personality traits the author attributes to Galileo. How are these similar or different from the other scientists you've read about?
• Galileo understood the need for "accurate, repeated experiments." Explain how his observation of the swinging chandelier helps him realize this need.

• **IMPORTANT DATES**
Galileo (1564-1642)

WEEK 4

⬚ **30m General Science: Grade 8 - Lesson 11**
Enter Galileo

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.5 p.66-71 "It is in Pisa" - "scientific questioning."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe Galileo's supposed experiment in Pisa.

★ TEACHER NOTE
The theological beliefs of Bruno are mentioned on p.71 within the context of heresy and politics. Teachers might choose to allow it to pass by the way or to engage students in discussion.

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• How is Galileo's personality getting in his way?
• A recurring theme, the authorities are very concerned about dangerous ideas. What are they afraid of?

WEEK 4

⬚ **30m General Science: Grade 8 - Lesson 12**
Galileo, the Thinker

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.6 p.72-79 "Galileo isn't" - "understand that."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how scientists, like Galileo, use math to understand relationships.
• What are mathematical relationships? (Mathematical ideas, like geometry and patterns, are important!)
• Tell how Galileo discovered his Law of Freefall.

→ SUPPLEMENTAL
∞ Video Link: Hammer Versus Feather on the Moon (1:22)

★ TEACHER TIP
Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

WEEK 4

⬚ **45m Labs: Grade 8 - Lesson 4**
Galileo's Falling Balls

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Galileo's Falling Balls, as directed in the Lab Book.

Suggested day 2: Students complete the Procedure this week.

WEEK 5

⬚ **30m Natural History: Grade 8 - Lesson 5**
Infectious Disease Activity

Materials: video link, cups prepared with milk and cornstarch (see Teacher Tip), iodine drops

PREP: Read Teacher Tip

→ INTRO
Recall what you have learned so far. Some microorganisms can make us sick. Understanding these microbes helps us protect ourselves and care for others. Watch the linked video to see one experiment. Viruses are specifically mentioned in the video, but the same idea applies to many disease-causing germs. Then you will do a similar experiment and compare your results.
∞ Video Link: How Viruses (and other germs) Spread (8:46)

→ ACTIVITY
• There are 8-12 cups. Each cup represents the 'body fluid' of a person. We will simulate a typical daily interaction between two people by swirling two cups, pouring the fluid first into one and then the other cup, and finally dividing it again between the two cups. Each person should 'interact' with 3-4 others.

• Complete the interactions for the group.

★ TEACHER TIP
Prep for this week's activity: a cup with about 2 Tbsp of milk should be prepared for each group member. If a student is working alone, then 8-12 cups could be prepared to simulate a group. Write names on the cups to differentiate. "Infect" one of the cups with 2 tsp of cornstarch and mix well.

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing how scientists have developed that knowledge? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

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Term 1

- Only one was originally infected. How many do you think contracted our pretend illness? Can you see any difference by looking at the cups? Add 1-2 drops of iodine to each cup and swirl one last time.

→ OBSERVE, NARRATE, & DISCUSS
- Each person whose 'body fluid' changed color was infected. What percentage of the total number were infected? How does this compare to what you thought would happen? Can you determine who infected the rest of the party?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 5

30m General Science: Grade 8 - Lesson 13
Thinking about Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.7 p.80-83 "Motion. How is" - "because it's important."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain Galileo's thought experiment and his Principle of Relativity.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 5

30m General Science: Grade 8 - Lesson 14
Thinking about Motion

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.7 p.83-89 "What Galileo does" - "at its core."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain how Galileo's Relativity applies to our observations, looking at the sky, a moving vehicle, or a thrown ball.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in motion? More interest/ease in discussing the relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a play ground, a science museum with an engineering exhibit, or an art museum with kinetic art. Reach out through Contact Us if you need help.

WEEK 5

30m General Science: Grade 8 - Lesson 15
Thinking about Stars

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.8 p.90-94 "Despite his enemies" - "feel the pull."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Contrast the reaction of most people when they see the supernova or the comet with Galileo's. How is this an echo of Galileo's larger problem with the authorities?
- Describe Galileo's viewpoint on revealed knowledge.

→ SUPPLEMENTAL

★ TEACHER NOTE
Galileo distinguishes between answers found through the use of our senses or in the Bible on p.92: "I do not" - "in the Bible." The age of the universe is mentioned throughout p.95-97. Teachers might choose to allow these to pass by the way, to discuss with students, or to consider alternative theories they would like to share. The optional reading could be skipped if preferred.

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Term 1

∞ Video Link: What is a Comet Made of? (3:31)
∞ Video Link: Night time Lapse of Comet Lovejoy (1:44)

→ NOTE
Read About Suns and Stuff p.95-97 for interest, if desired.
• Describe or diagram the life cycle of a star, including descriptions of each phase.

WEEK 5

45m Labs: Grade 8 - Lesson 5
Galileo's Falling Balls

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of Galileo's Falling Balls, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Analysis and Conclusions this week.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 6

30m Natural History: Grade 8 - Lesson 6
Meet TB

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ INTRO
Look at the title and cover. Have you ever heard of tuberculosis (TB)? TB is one of those disease-causing germs. This story is about its part in natural history.

→ READ, NARRATE, & DISCUSS
Ch.1 p.1-7 "This is" - "they encountered."

• On p.4, the author says, "The germs began to evolve. What emerged was a different and even more dangerous microorganism…" Discuss how this happened in terms of "natural selection"?
∞ Video Link: Natural Selection (5:02)

★ TEACHER TIP
There is an INTRO for this week's lesson, but plan to wean students from these prompts and train them to RECAP on their own. This will smooth their transition into high school coursework.

★ TEACHER NOTE
The age of the Earth and evolution are mentioned multiple times over p.1-6 and in the supplemental video. Teachers should preread the lesson if they have concerns and consider how they would like to discuss or what other theories they might like to share.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 6

30m General Science: Grade 8 - Lesson 16
Galileo Challenges the World

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.9 p.98-105 "Galileo is" - "the heavens go."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What happens when Galileo gets a telescope?

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• IMPORTANT DATES
Invention of the telescope (1609)

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Term 1

	• How does Galileo's personality contribute to his conflict with authority? **➔ SUPPLEMENTAL** ∞ Video Link: Survive a Trip to the Sun (3:20)	
WEEK 6	⬚ **30m General Science: Grade 8 - Lesson 17** *The Telescope* Materials: Newton at the Center **➔ RECAP** Briefly recall what you read last time. **➔ READ, NARRATE, & DISCUSS** Ch.9 p.106-109 "Leonard Digges" - "in the shadows." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Who invented the telescope?	★ TEACHER NOTE The scale of the universe in millions of light-years is mentioned on p.108. Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.
WEEK 6	⬚ **30m General Science: Grade 8 - Lesson 18** *Galileo in Trouble* Materials: Newton at the Center PREP: Read Teacher Tip **➔ RECAP** Briefly recall what you read last time. **➔ READ & NARRATE** Ch.10 p.110-115 "Galileo doesn't" - "of his time." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. **➔ READ, NARRATE, & DISCUSS** Ch.10 p.116-117 "Religion is" - "search a bit." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • We see religious leaders both angry with and engaging with Galileo. Why were they ultimately upset with Galileo? How could the situation have been handled differently? • Explain religion and science using a metaphor of 2 parallel lines of thinking in the mind. • Is the title of the passage on p.116-117 the best title for the ideas presented? If so, explain. If not, what would you title it?	★ TEACHER TIP When passages have different main topics, some students will be able to read them through and organize the content in their minds. For other students, it can be helpful to read and narrate the passages separately, even if they are doing the work consecutively in the same lesson period. This helps the mind work through the ideas in smaller chunks. ★ TEACHER NOTE The balance between doctrine and politics is considered on p.113, 116-117. Teachers are encouraged to preread these sidebars for better discussion.
WEEK 6	⬚ **45m Labs: Grade 8 - Lesson 6** *Magic of Inertia Activity* Materials: Grade 8 Lab Book and materials listed within PREP: Read Teacher Tip **➔ LAB DAY** Complete the Magic of Inertia activity, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.	★ TEACHER TIP This is a brief activity that will be completed this week. Be sure to have them share their notebook, during which they should be able to discuss the science and explain what they did.

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WEEK 7

⬚ **30m Natural History: Grade 8 - Lesson 7**
TB in the Ancient World

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.1 p.8-13 "The existence" - "throughout Europe."

★ TEACHER NOTE
The age of the civilization is mentioned on p.8: "hundreds of thousands of years." Teachers might choose to allow it to pass by the way or to consider how they would like to discuss or offer alternative theories they would like to share.

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 7

⬚ **30m General Science: Grade 8 - Lesson 19**
Kepler's Theories on Planetary Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.11 p.118-124 "Do you think" - "published in 1611."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Tell about Kepler.
● Describe Keplar's interest in optics and how this leads to a connection with Galileo.

● **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

● **IMPORTANT DATES**
Johannes Kepler (1571-1630)

WEEK 7

⬚ **30m General Science: Grade 8 - Lesson 20**
Kepler's Theories on Planetary Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.11 p.124-128 "Meanwhile" - "do just that"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.11 p.129-131 "An ellipse" - "Earth and Sun."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain Kepler's belief that his work as a scientist is God's intent for him, his vocation.
● Contrast Kepler's thoughts on creation with those of previous scientists.
● How does Kepler's work generate new questions?
● What is an ellipse? What is eccentricity?

★ TEACHER NOTE
The timescale of change in Earth's orbital path is mentioned on p.131: "in a 100,000-year cycle." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

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Science: Grade 8

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WEEK 7

⬚ **30m General Science: Grade 8 - Lesson 21**
Descartes and His Coordinates

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.12 p.132-136 "Galileo is 32" - "those coordinates."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain how the Cartesian coordinate system works.

● IMPORTANT DATES
René Descartes (1596-1650)

WEEK 7

⬚ **45m Labs: Grade 8 - Lesson 7**
Forces

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Forces, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. If time permits, they might complete step 1 of the Procedure.

∞ Video Link: Step 1b How-to (2:32)

★ TEACHER TIP
Students adjust variables to the ramp from Lab 2 to evaluate forces related to gravity, friction, etc. Expect students to refer back to the process of using a spreadsheet from Lab 2. Reaching back to apply what was previously learned in a new situation is important. Encourage and model for them.

WEEK 8

⬚ **30m Natural History: Grade 8 - Lesson 8**
Medieval Medicine

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.2 p.14-17 "For days" - "human urine."

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m General Science: Grade 8 - Lesson 22**
Descartes and His Coordinates

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.12 p.136-139 "Like him or not" - "home to rest."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.12 p.141-143 "On June 21" - "hundreds of years."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

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● Explain how even wrong ideas, such as Descartes' vortices, can be helpful in science.
● Tell the story of Fermat's Last Theorem.

WEEK 8

30m General Science: Grade 8 - Lesson 23
Meet Isaac Newton

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.13 p.144-147 "It turns out" - "ever happening."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Tell about Newton's early life.

● IMPORTANT DATES
Isaac Newton (1642-1727)

WEEK 8

30m General Science: Grade 8 - Lesson 24
Newton Ponders Motion and Math

Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.13 p.147-153 "When he is 16" - "he is doing."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Describe Newton's personality traits. How does he relate to others? What kinds of things does he think about?
● Tell the story of how Newton learns math and invents calculus.
● Explain the links in the chain that led to Newton's ideas on universal gravity.

WEEK 8

45m Labs: Grade 8 - Lesson 8
Forces

Materials: Grade 8 Lab Book and materials listed within

➔ LAB DAY
Complete day 2 of Forces, as directed in the Lab Book.

Suggested day 2: Students work through the Procedure today. If they need to take a break and complete the Procedure next week, they should do so between steps as directed in the lab.

WEEK 9

30m Natural History: Grade 8 - Lesson 9
Renaissance Medicine

Materials: Invincible Microbe

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.2 p.17-22 "While superstition" - "to get worse."

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

● IMPORTANT DATES
Thermometer invented (1715)
Stethoscope invented (1818)

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WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 25**
Newton Puzzles Over Gravity

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.14 p.154-159 "The celebrated" - "that it doesn't."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Why is this chapter so titled?

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 26**
What is Calculus?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.14 p.160-163 "With arithmetic" - "at any point."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain what calculus is and what it can be used for.

WEEK 9 ⬚ **30m General Science: Grade 8 - Lesson 27**
Thinking about Light

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.15 p.164-171 "Newton keeps" - "that new world."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Discuss Newton's work related to light.
- Discuss Newton and Hooke's disagreement and their relationship.
- Discuss Newton's concerns about religion.

WEEK 9 ⬚ **45m Labs: Grade 8 - Lesson 9**
Forces

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of Forces, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: If students have not yet done so, they should complete

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

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the Procedure. Analysis and Conclusions today. If they need an extra week, take the time needed to finish. There should be plenty of time for the last activity in Week 11.

WEEK 10 ☐ **30m Natural History: Grade 8 - Lesson 10**

Industrial Surge

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.3 p.23-30 "It was a" - "mumbo jumbo."

• The author states that the Industrial Revolution caused tuberculosis to spread. Explain.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ☐ **30m General Science: Grade 8 - Lesson 28**

Newton's Laws of Motion

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.16 p.172-177 "Is motion" - "in his head."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain Newton's Laws of Motion.
• Discuss the implications of these laws in relation to astronomy.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 10 ☐ **30m General Science: Grade 8 - Lesson 29**

Fame Finds Newton

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.17 p.178-183 "Newton has" - "himself a star."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss Newton and Halley's relationship. What impact does it have on Newton?
• Why was the Principia important?

• IMPORTANT DATES
Newton publishes Principia (1687)

WEEK 10 ☐ **30m General Science: Grade 8 - Lesson 30**

Fame Finds Newton

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.17 p.184-187 "The French had" - "last by science."
Note ideas that stand out to you in words, pictures, or diagrams. Record

• IMPORTANT DATES
Halley's Comet (1758)
Appeared last (1986)
Next appearance (2061)

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any questions or connections you make along the way, too.

• Describe the appearance of Halley's Comet from the perspective of someone in 1758. What do you imagine they were thinking?

WEEK 10 ⬚ **45m Labs: Grade 8 - Lesson 10**
Equal and Opposite Activity

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of the Equal and Opposite activity, as directed in the Lab Book. Students may complete this activity in 1 week, but may also spend more time.

★ TEACHER TIP
Students play with balls to observe Newton's 3rd law.

WEEK 11 ⬚ **30m Natural History: Grade 8 - Lesson 11**
Catch Up

▯ Materials: Invincible Microbe

PREP: Read Teacher Tip

➔ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 31**
Measuring Light

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.18 p.188-195 "Can the speed" - "at the sky."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell the story of Roemer and Jupiter's moons.
• Consider the significance of Roemer's realization.

• IMPORTANT DATES
Olaus Christensen Roemer (1644-1710)
Christiaan Huygens (1629-1695)

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 32**
Measuring Light

▯ Materials: Newton at the Center

PREP: Read Teacher Tip

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether motion or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more

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Science: Grade 8

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Term 1

	Ch.18 p.196-197 "Of the four" - "more impressive." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Study the diagram on p.197. Why would Jupiter's light be brighter sometimes than at other times? If one year equals one trip around the sun, how old would you be if you lived on Jupiter?	reading support, less business, etc? Reach out through Contact Us if you need help.
WEEK 11	⬚ **30m General Science: Grade 8 - Lesson 33** *Catch Up* Materials: Newton at the Center **→ CATCH UP** Use this time to catch up on lessons in this or another course, or explore Extra Helpings.	
WEEK 11	⬚ **45m Labs: Grade 8 - Lesson 11** *Equal and Opposite Activity* Materials: Grade 8 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete the Equal and Opposite activity or use this time to catch up on other labs or lessons in this course. If you have not already, be sure to narrate the lab to your teacher by showing them your lab notebook.	★ TEACHER TIP Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.
WEEK 12	⬚ **30m Natural History: Grade 8 - Lesson 12** *Exam Week* **→ EXAMS** Answer question(s) related to course.	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 34** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 35** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	

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Term 1

WEEK 12 ⬚ **45m Labs: Grade 8 - Lesson 12**

➔ EXAMS
Answer question(s) related to course.

Questions will come from:
Grade 8 Lab Book

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Science: Grade 8

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Term 2

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 13**
The Sanatorium

▯ Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.31-33 "Hermann Brehmer" - "when forced to."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 37**
Thinking about Matter

▯ Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.19 p.198-203 "Aristotle thought" - "close to finding it."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Set the stage for alchemy.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 38**
Thinking about Matter

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.19 p.203-207 "Bernard of Treves" - "resounding 'Yes.'"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.19 p.208-209 "Hennig Brandt" - "specific discoverer."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about the demise of alchemy.
• Tell about the discovery of phosphorus.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 39**
Measuring Matter

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.20 p.210-215 "Irish-born Robert" - "corpuscles seriously."
Note ideas that stand out to you in words, pictures, or diagrams. Record

• IMPORTANT DATES
Robert Boyle (1627-1691)

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Term 2

any questions or connections you make along the way, too.

• Tell about Robert Boyle.

WEEK 1

45m Labs: Grade 8 - Lesson 13
Under Pressure Activity

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete the Under Pressure activity, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

★ TEACHER TIP
Students evaluate pressure and volume in 3 common phases of matter. Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 2

30m Natural History: Grade 8 - Lesson 14
The Sanatorium

Materials: Invincible Microbe

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.4 p.35-39 "Some sanatoriums" - "hope of a cure."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

30m General Science: Grade 8 - Lesson 40
Is Air Something - or Nothing?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.20 p.216-219 "Air seems" - "popular entertainment."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Is air something or nothing?

WEEK 2

30m General Science: Grade 8 - Lesson 41
Thinking about Energy and Matter

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.21 p.220-225 "Daniel Bernoulli" - "ahead of the times."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about the Bernoulli brothers.

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WEEK 2	⬚ **30m General Science: Grade 8 - Lesson 42** *Thinking about Energy and Matter* Materials: Newton at the Center **→ RECAP** Briefly recall what you read last time. **→ READ, NARRATE, & DISCUSS** Ch.21 p.225-229 "Now, let's skip" - "to technology." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Explain or diagram what you understand about Daniel Bernoulli's principle.	**• IMPORTANT DATES** Daniel Bernoulli (1700-1782)
WEEK 2	⬚ **45m Labs: Grade 8 - Lesson 14** *A Degree of Challenge* Materials: Grade 8 Lab Book and materials listed within PREP: Read Teacher Tip **→ LAB DAY** Complete day 1 of A Degree of Challenge, as directed in the Lab Book. Suggested day 1: Students complete the Introduction. If time permits, they may complete the Procedure up to step 2.	★ TEACHER TIP Students test the relationship between volume and pressure. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.
WEEK 3	⬚ **30m Natural History: Grade 8 - Lesson 15** *Patients are Persons* Materials: Invincible Microbe **→ RECAP** Briefly recall what you read last time. **→ READ, NARRATE, & DISCUSS** Ch.5 p.40-44 "In 1937" - "lot of fun." • Do you think the patients were people who usually behaved as described in the last paragraph of the reading? Explain their behavior.	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 3	⬚ **30m General Science: Grade 8 - Lesson 43** *"Nature is squaring itself."* Materials: Newton at the Center **→ RECAP** Briefly recall what you read last time. **→ READ, NARRATE, & DISCUSS** Ch.22 p.230-237 "Think gorgeous" - "built on it." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Tell why Émilie du Châtelet didn't fit in. • Explain the mistake that Émilie discovered.	★ TEACHER NOTE The affair between Émilie du Châtelet and Voltaire is introduced on p.232. Teachers might choose to allow it to pass by the way or to discuss with students. **• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link. **• IMPORTANT DATES** Emilie du Châtelet (1706-1749) Voltaire (1694-1778)

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WEEK 3

⬚ **30m General Science: Grade 8 - Lesson 44**
Many Gases

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.23 p.238-245 "Just what is" - "have them?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Chemistry is just getting started, but it is very disorganized. Discuss.

★ TEACHER NOTE
The age of some rock is mentioned in a caption on p.239, "roughly 65 million years or so ago." Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• IMPORTANT DATES
Joseph Priestley (1733-1804)
Benjamin Franklin (1706-1790)

WEEK 3

⬚ **30m General Science: Grade 8 - Lesson 45**
The Thermometer

▯ Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.23 p.246-249 "Daniel Fahrenheit" - "the temperature."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell about Daniel Fahrenheit.

• IMPORTANT DATES
Daniel Fahrenheit (1686-1736)

WEEK 3

⬚ **45m Labs: Grade 8 - Lesson 15**
A Degree of Challenge

▯ Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of A Degree of Challenge, as directed in the Lab Book.

Suggested day 2: Students work on the Procedure this week. If step 2 was completed last week, then they might also complete Analysis and Conclusions. If they are just beginning the Procedure this week, then they may take another week to finish the lab.

WEEK 4

⬚ **30m Natural History: Grade 8 - Lesson 16**
Patients are Persons

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.5 p.44-50 "Of course" - "cure--forever."

• Describe a typical day in the life of a patient in a sanatorium. What do you think about this?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 46**
Weighing the World

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.24 p.250-259 "Henry Cavendish" - "Earth's density."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how Cavendish measured the density of the world using words, drawings, or diagrams.

★ TEACHER NOTE
The Big Bang is mentioned toward the end of p.258. Teachers might choose to allow it to pass by the way or to consider alternative theories they would like to share.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Henry Cavendish (1731-1810)

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 47**
Chemistry is Finally Here!

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.25 p.260-265 "It is the" - "revolution in science."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss what the "big idea" is in this chapter. What is Lavoisier doing?

• IMPORTANT DATES
Antoine-Laurent Lavoisier (1743-1794)

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 48**
Lavoisier Establishes Chemistry

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.26 p.266-274 "His parents" - "ideas survive."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Why is Lavoisier called the "Father of Chemistry"?
• What is an element?
• What do we mean by "conservation of mass"?
• Why is hypothesizing an important part of the scientific method?

➔ NOTE
Read France Sings a Metric Tune p.275-277 for interest, if desired.

WEEK 4 ⬚ **45m Labs: Grade 8 - Lesson 16**
A Degree of Challenge

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 3 of A Degree of Challenge, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

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notebook.

Suggested day 3: Students complete the lab this week. Some students who finished last week might jump ahead to the next lab.

WEEK 5

⬚ **30m Natural History: Grade 8 - Lesson 17**

The Cause

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.6 p.51-55 "Robert Koch" - "economic stability."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing the relationship between infectious disease and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• IMPORTANT DATES
Robert Koch identifies TB (1882)

WEEK 5

⬚ **30m General Science: Grade 8 - Lesson 49**

Elements and Atoms

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.27 p.278-287 "Way back" - "explain their world."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss how Dalton builds on the ideas of Lavoisier.
• Create a graphic or diagram of Dalton's ideas about elements and atoms.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
John Dalton (1766-1844)

WEEK 5

⬚ **30m General Science: Grade 8 - Lesson 50**

Atoms and Molecules

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.28 p.288-293 "Amedeo Avogadro" - "complicated things."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.28 p.294-297 "So what makes" - "New Dimension."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

★ TEACHER TIP
Are your learner(s) demonstrating more familiarity/interest in chemistry or matter? More interest/ease in discussing the relationship between science and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a chemistry exhibit. Reach out through Contact Us if you need help.

• IMPORTANT DATES
Amedeo Avogadro (1776-1856)

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- Explain atoms versus molecules.
- Explain Avogadro's Law.
- Discuss how difficult it can be to believe something you cannot see. Consider how people of science and religion both experience this.
- Tell how and what Edward Frankland discovered about chemical bonding.

WEEK 5

30m General Science: Grade 8 - Lesson 51

Making Sense of Matter

Materials: Newton at the Center

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.29 p.298-308 "Scientists don't" - "that's exciting."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Mendeleyev's life.
- Tell about Mendeleyev's work.

• **IMPORTANT DATES**
Dmitri Mendeleyev (1834-1907)
Periodic Table of Elements (1869)

WEEK 5

45m Labs: Grade 8 - Lesson 17

Elements and Reactions

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

Complete day 1 of Elements and Reactions, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure this week. There will be a wait of at least 1 week following step 7.

★ TEACHER TIP
Students evaluate the reaction of iron and oxygen when steel wool rusts.

WEEK 6

30m Natural History: Grade 8 - Lesson 18

Fighting the Germ

Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS

Ch.6 p.55-60 "Mustering support" - "newfangled machine."

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• **IMPORTANT DATES**
X-Rays discovered (1895)
Hermann Biggs (1859-1923)

WEEK 6

30m General Science: Grade 8 - Lesson 52

The Periodic Table

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.29 p.309-311 "So far" - "New Dimension."
Note ideas that stand out to you in words, pictures, or diagrams. Record

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• **IMPORTANT DATES**
The four newest elements added to the Periodic Table (2019)

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any questions or connections you make along the way, too.

- Tell what you know about the Periodic Table.

WEEK 6

☐ **30m General Science: Grade 8 - Lesson 53**
Thinking about Heat

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.30 p.312-321 "Fill a teacup" - "two philosophers!"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Benjamin Thompson's life.
- Explain Thompson's ideas about heat.

- **IMPORTANT DATES**
Benjamin Thompson (1753-1814)

WEEK 6

☐ **30m General Science: Grade 8 - Lesson 54**
Studying Electricity

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.31 p.322-327 "You may remember" - "Newton's ideas."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell a story about the Leyden jar.
- How does Benjamin Franklin "link action in the heavens to that on Earth"?

- **IMPORTANT DATES**
Benjamin Franklin (1706-1790)

WEEK 6

☐ **45m Labs: Grade 8 - Lesson 18**
Elements and Reactions

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Elements and Reactions, as directed in the Lab Book.

Suggested day 2: Students continue the Procedure.

WEEK 7

☐ **30m Natural History: Grade 8 - Lesson 19**
Fighting the Germ

Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.6 p.60-64 "The influence" - "called outsiders."

- **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 7	⬚ **30m General Science: Grade 8 - Lesson 55** *Studying Electricity* Materials: Newton at the Center **➔ RECAP** Briefly recall what you read last time. **➔ READ, NARRATE, & DISCUSS** Ch.31 p.327-333 "Franklin isn't" - "path of science." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Discuss how Volta harnesses electricity. • Explain what is learned about elements and magnetism through this harnessed electricity.	**• NATURE NOTEBOOK** Prompt for General Science in Outdoor Work Quick Link. **• IMPORTANT DATES** Alessandro Volta (1745-1827)
WEEK 7	⬚ **30m General Science: Grade 8 - Lesson 56** *A Pendulum's Proof* Materials: Newton at the Center **➔ RECAP** Briefly recall what you read last time. **➔ READ, NARRATE, & DISCUSS** Ch.31 p.334-337 "Way back" - "Amir D. Aczel." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Describe Foucault's pendulum experiment. **➔ SUPPLEMENTAL** ∞ Video Link: Foucault's Pendulum Explained (3:20) ∞ Video Link: Foucault Pendulum Time Lapse (watch to 4:45)	**• IMPORTANT DATES** Jean-Bernard-Leon Foucault (1819-1868) Foucault's Pendulum Experiment (1851)
WEEK 7	⬚ **30m General Science: Grade 8 - Lesson 57** *Faraday Experiments* Materials: Newton at the Center **➔ RECAP** Briefly recall what you read last time. **➔ READ, NARRATE, & DISCUSS** Ch.32 p.338-345 "It is January" - "he isn't finished." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Explain Faraday's concept of "fields." • Tell about Faraday's efforts to improve communication in science. Why is communication so important? • Discuss Faraday's habit of finding an opportunity to learn in every situation, that is, learning for the sake of learning. • Explain Faraday's hypothesis and experiment about electricity and magnetism.	**• GEOGRAPHY WORK** Michael Faraday (1791-1867)

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WEEK 7

⬚ **45m Labs: Grade 8 - Lesson 19**
Elements and Reactions

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of Elements and Reactions, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 8

⬚ **30m Natural History: Grade 8 - Lesson 20**
The Outsiders

Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.7 p.65-70 "In November 1906" - "beds available."

- Explain how AMA practices and policy kept people of color out of medicine. What effect did this have on communities?

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

⬚ **30m General Science: Grade 8 - Lesson 58**
Relating Electricity and Magnetism

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32 p.345-351 "Faraday is" - "Victoria."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- How was imagination important to Faraday's work?
- Explain how Faraday imagined light in his fields.

● **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8

⬚ **30m General Science: Grade 8 - Lesson 59**
Thinking about Light

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32 p.352-355 "The Bible says" - "an understatement."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Is light a particle or a wave?

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WEEK 8

⬚ **30m General Science: Grade 8 - Lesson 60**
Maxwell's Equations

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.33 p.356-365 "James Clerk Maxwell" - "of all time."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Maxwell's life.
- Explain what Maxwell does with Faraday's ideas on light. How do the two scientists complement each other?
- "Today, James Clerk Maxwell is ranked with the greatest scientists of all time." Tell why this is, using any combination of pictures and/or words.

• IMPORTANT DATES
James Clerk Maxwell (1831-1879)

WEEK 8

⬚ **45m Labs: Grade 8 - Lesson 20**
Moving Particles, Electrical Energy, Magnetic Force

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of Moving Particles, Electrical Energy, and Magnetic Force, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure this week.

★ TEACHER TIP
Students experiment with wire to observe a relationship between electricity and magnetism.

WEEK 9

⬚ **30m Natural History: Grade 8 - Lesson 21**
The Outsiders

▯ Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.7 p.70-75 "From the late" - "cooks and drivers."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

• IMPORTANT DATES
The Great Migration (1915-1930)

WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 61**
What is Frequency?

▯ Materials: Newton at the Center

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.33 p.366-369 "My kids tell"- "than light."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain frequency, using any combination of pictures and/or words.

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WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 62**
What is Heat?

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.34 p.370-373, 377-379 "Ludwig Eduard"- "related suicide."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Boltzmann's life.
- Describe how Boltzmann built upon Maxwell's ideas.
- Explain the frustrating debate about atoms.

★ TEACHER NOTE
The suicide of Boltzmann is mentioned in the last sentence on p.379. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Ludwig Boltzmann (1844-1906)

WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 63**
Invisible Ideas

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ & NARRATE
Ch.34 p.374-376 "Gas molecules" - "and direction."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

→ READ, NARRATE, & DISCUSS
Ch.34 p.380-383 "In this book" - "managed it."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain the relationship between kinetic energy and temperature.
- Summarize what you know about atoms.

WEEK 9

⬚ **45m Labs: Grade 8 - Lesson 21**
Moving Particles, Electrical Energy, Magnetic Force

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 2 of Moving Particles, Electrical Energy, and Magnetic Force, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 10

⬚ **30m Natural History: Grade 8 - Lesson 22**
The Outsiders

Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.7 p.75-81 "Edythe Tate-Thompson" - "as outsiders."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

• When you are reading, do you sympathize more easily with those who are sick with TB or those who are afraid of getting sick? Try to imagine you are part of whichever group you did NOT choose. Now explain the problems with which you are faced as part of that other group. How do you feel about those problems? What are some reasonable solutions to those problems?

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 64**

Putting Energy to Work

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.35 p.384-389 "This is a" - "to human effort."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What is work?
• Explain or draw a series of diagrams that show the potential and kinetic energy of a ball being thrown up into the air.

→ NOTE
Read Information-Age Thinking in the Industrial Era p.390-393 for interest, if desired.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
James Joule (1818-1889)

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 65**

First Law of Thermodynamics

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.36 p.394-399 "We have now"- "discoveries of physics."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain "conservation of energy."

• IMPORTANT DATES
Julius von Mayer (1814-1878)
Ferdinand von Helmholtz (1821-1894)

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 66**

Second Law of Thermodynamics

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37 p.400-405 "The first law" - "be used again."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Entropy can be tricky to understand. Explain it in terms of "energy spread."

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Term 2

WEEK 10 ⬚ **45m Labs: Grade 8 - Lesson 22**

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of Faraday's Discovery, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction and begin the Procedure.

★ TEACHER TIP
Students produce electricity using a magnetic field.

WEEK 11 ⬚ **30m Natural History: Grade 8 - Lesson 23**
Catch Up

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ CATCH UP
Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

★ TEACHER TIP
Next term begins with an activity! If needed, plan to prepare agar plates according to the package directions.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 67**
Second Law of Thermodynamics

Materials: Newton at the Center

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37 p.405-411 "Now if I've" - "a meaningful answer."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain entropy in terms of probability.
● Is there a difference between understanding entropy in terms of energy spread or probability?

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether chemistry, matter, electricity, magnetism, or the relationship between science and society are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help.

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Term 2

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 68**
Things We Don't Understand

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.38 p.412-415 "If you want" - "a Nobel Prize."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- What does imagination have to do with science?

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 69**
Pursuit of Scientific Truth

Materials: Newton at the Center

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.39-40 p.416-431 "Albert Michelson" - "of all time."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- How would you answer someone today who argues that everything there is to know about a scientific field has already been discovered?
- What is more important in science: theory or experimentation?

• IMPORTANT DATES
Antoine-Henri Becquerel (1852-1908)
First X-Rays (1895)

WEEK 11 ⬚ **45m Labs: Grade 8 - Lesson 23**
Faraday's Discovery

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 2 of Faraday's Discovery, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 12 ⬚ **30m Natural History: Grade 8 - Lesson 24**
Exam Week

PREP: Read Teacher Tip

→ EXAMS
Answer question(s) related to course.

★ TEACHER TIP
Next term begins with an activity! If needed, plan to prepare agar plates according to the package directions.
∞ **Video Link:** Pouring Agar Plates (2:51)

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 2

WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 70** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 71** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 72** *Exam Week* **➔ EXAMS** Answer question(s) related to the course. Questions will come from: Newton at the Center	
WEEK 12	⬚ **45m Labs: Grade 8 - Lesson 24** **➔ EXAMS** Answer question(s) related to course. Questions will come from: Grade 8 Lab Book	

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Term 3

WEEK 1

⬚ **30m Natural History: Grade 8 - Lesson 25**

Antimicrobial Activity

▯ Materials: three antimicrobial agents used at home, three agar plates, permanent marker, sensitivity discs, small forceps/tweezers, gloves, cotton swabs, rubbing alcohol or hot soapy water

➔ INTRO
Recall what is happening in the story of TB. Think of three cleaning agents or natural remedies you might use to get rid of germs. How well do they work? Let's find out in this activity.

➔ ACTIVITY
• Turn all of the agar plates upside down and use a marker to divide them into quadrants. Write "Control" in one quadrant of each plate and label each of the remaining quadrants for each of the antimicrobial agents to be tested. So, plate 1 will have the control, and conditions A, B, and C in the 4 quadrants. Plates 2 and 3 will repeat the same sample set. When the plates are all labeled, turn them back over.

• With gloved hands, use a sterile cotton swab to rub the inside of your mouth for 10 seconds. Lift the lid of one plate just enough to evenly and gently rub the cotton swab over the entire surface. Then, recover the plate with the lid. Repeat for all three plates.

• Use rubbing alcohol to sterilize the forceps or wash them in hot, soapy water. Then pick up a single, sensitivity disc with the forceps and apply the first of your antimicrobial agents to the disc. Remove the lid of one plate and place the disc into the center of the appropriate quadrant. Press gently with the forceps, but do not damage the agar growth medium. Repeat for plates 2 and 3.

• Clean the forceps again and then repeat the last step for the 2 remaining antimicrobial agents and the control. The control will be a plain disc.

• When you are finished, turn all the plates back upside down and incubate in a warm (but not hot) location. You may wish to tape them closed. If you have time to observe them during the week, then do. If not, then they can wait until next week.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 73**

Introducing Einstein and Recapping

▯ Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

➔ INTRO
This book continues the story of science from Newton at the Center. We will learn how the ideas and culture of science developed in surprising directions in the modern era, even as the process of science remained the same. These concepts are challenging to grasp and are meant to stretch the mind and engage the imagination. It is not essential that every technical concept be fully understood. Let the lab book guide you on which concepts to focus on. Break the assigned reading into chunks, as needed. Reading and narrating smaller chunks can be helpful even when reading in one sitting. Allowing the reading to spill over into free reading time is also okay. Learning what works for you is an important skill that will help in high school.

★ TEACHER TIP
Passages are not presented in parts this term. Observe how students manage them and encourage them to read and narrate the ideas in chunks, as needed.

★ TEACHER NOTE
A review of religious and political conflict is covered on p.10-12. Teachers might choose to allow it to pass by the way or to discuss with students.

• IMPORTANT DATES
Albert Einstein (1879-1955)
Industrial Revolution (1760-1840)
Euclid (4th B.C.)

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Science: Grade 8

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Term 3

→ **READ, NARRATE, & DISCUSS**
Ch.1-2 p.1-17 "Fifteen-year-old" - "ever seeing one?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Discuss why Albert Einstein had such a hard time in school.
- Consider what it would be like if you could travel on a beam of light.
- Recall some of the historical events in science that have taken place so far. Recall some of the "big ideas" we've encountered.

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 74**
Electricity, Magnetism, and Light

Materials: Einstein Adds a New Dimension

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.3-4 p.18-35 "Science has" - "method employed."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Science is the search for connections and patterns. Discuss this statement.
- Describe Michelson and Morley's experiment.

• **IMPORTANT DATES**
Samuel Morse (1791-1872) sent an electric current from Washington D.C. to Baltimore (1844).
Thomas Edison (1847-1931)
Nikola Tesla (1856-1943)
Any dates from the timeline on p.20-21

WEEK 1

⬚ **30m General Science: Grade 8 - Lesson 75**
Thinking about Electrons

Materials: Einstein Adds a New Dimension

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.5 p.36-49 "Joseph John Thomson" - "skilled experimenter."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Describe the debate surrounding electricity.

• **IMPORTANT DATES**
J.J. Thomson (1856-1940) presents his corpuscles of electricity to the Royal Institution (1897).
Max Planck (1858-1947)
Robert Millikan (1869-1953)

WEEK 1

⬚ **45m Labs: Grade 8 - Lesson 25**
The Photoelectric Effect

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ **LAB DAY**
Complete day 1 of The Photoelectric Effect, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

★ TEACHER TIP
Students build a device to detect electric charge and observe how light interacts. Remember the most important objective is always building skills and habits. Pacing is only a suggestion. Students should engage with lab at a pace that is appropriate for their abilities and interest.

WEEK 2

⬚ **30m Natural History: Grade 8 - Lesson 26**
Antimicrobial Activity

Materials: plates from last week

→ **INTRO**
Today, you will examine your plates! How well do you think your antimicrobial agents worked?

• **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Science: Grade 8

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Term 3

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your three plates and examine them. Draw or diagram them in your science notebook. Did your antimicrobial agents inhibit growth better than the control? Which of your agents worked the best? Does this mean that it works as well against all microbes? How could you find out?

WEEK 2

30m General Science: Grade 8 - Lesson 76
What are Atoms Anyway?

Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.6-7 p.50-57 "J.J. has two" - "the uranium."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What do scientists think about atoms at this time?
• Tell what you know about Marie Curie's work.

★ TEACHER TIP
Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries.

• **NATURE NOTEBOOK**
Prompt for General Science in Outdoor Work Quick Link.

• **IMPORTANT DATES**
Marie Curie (1867-1934)
Any dates from the timeline on p.57

WEEK 2

30m General Science: Grade 8 - Lesson 77
Investigating Atoms

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.8 p.58-69 "J.J. Thomson's" - "inside the atom."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Explain how work with radioactivity reveals an understanding of atoms.

• **IMPORTANT DATES**
Marie and Pierre Curie discovered polonium (1898).
Marie Curie isolated radium (1902).
Chernobyl Nuclear Power Plant explosion (1986)

WEEK 2

30m General Science: Grade 8 - Lesson 78
Theorizing

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.9-10 p.70-85 "Max Karl Ernst" - "is turned down."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• If Planck didn't believe his equation, why did he develop it? Why did he even think about it if he wasn't sure it was real?
• Discuss what Einstein did with all of the seemingly contradictory information.

★ TEACHER NOTE
A daughter born to Albert Einstein and Mileva Maric before their marriage is mentioned on p.80. Teachers might choose to allow it to pass by the way or to discuss with students.

• **IMPORTANT DATES**
Planck's constant quantizes energy (1900)
Einstein published his 5 famous papers (1905).

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Term 3

WEEK 2 ⬚ **45m Labs: Grade 8 - Lesson 26**
The Photoelectric Effect

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ **LAB DAY**
Complete day 2 of The Photoelectric Effect, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure and Analysis and Conclusions today.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 3 ⬚ **30m Natural History: Grade 8 - Lesson 27**
The 'Cure'

Materials: Invincible Microbe

→ **RECAP**

→ **READ, NARRATE, & DISCUSS**
Ch.8 p.82-86 "After announcing" - "drastic indeed."

● **OUTDOOR WORK**
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 79**
Light as Wave and Particle

Materials: Einstein Adds a New Dimension

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.11 p.86-97 "For 10 years" - "to the explanation."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Light is both a wave and a particle. Discuss.

● **IMPORTANT DATES**
Lord Rayleigh (1842-1919) explains why the sky is blue (1871).

WEEK 3 ⬚ **30m General Science: Grade 8 - Lesson 80**
Modeling the Unseen

Materials: Einstein Adds a New Dimension

→ **RECAP**
Briefly recall what you read last time.

→ **READ, NARRATE, & DISCUSS**
Ch.12-13 p.98-113 "It is 1827" - "to find out."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Describe how the role of mathematics in science is developing.
● Describe Rutherford's gold foil experiment.

● **IMPORTANT DATES**
Ernest Rutherford (1871-1937) conducted his gold foil experiment, revealing the nucleus of the atom (1909).

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Term 3

WEEK 3

⬚ **30m General Science: Grade 8 - Lesson 81**
Getting Atoms

▯ Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.14 p.114-127 "Rutherford and" - "astrophysics soared."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain how the concept of the atom changed from Thomson to Rutherford to Bohr.
● Explain Bohr and Einstein's difference of opinion. How can they be such good friends when they disagree so fundamentally?

★ TEACHER NOTE
The birth of the universe is mentioned in the sidebar on p.125. Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they want to share.

● IMPORTANT DATES
Niels Bohr (1885-1962) introduced his model of the atom (1913).

WEEK 3

⬚ **45m Labs: Grade 8 - Lesson 27**
Modeling Atoms

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 1 of Modeling Atoms, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction.

★ TEACHER TIP
Students create a model to convey what they know about the atom. Teachers should verify that students have what they need to begin their model next week.

WEEK 4

⬚ **30m Natural History: Grade 8 - Lesson 28**
The 'Cure'

▯ Materials: Invincible Microbe

➔ RECAP

➔ READ, NARRATE, & DISCUSS
Ch.8 p.86-91 "In Spain" - "doctor-approved cures."

● Describe some of the ineffective "cures" that were sold. What are the problems with these "cures"?

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 4

⬚ **30m General Science: Grade 8 - Lesson 82**
Still Shooting Alpha Particles

▯ Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.15 p.128-135 "In 1916" - "still ahead."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain what nuclear chemistry continues to reveal about the atom.

★ TEACHER NOTE
Carbon-14 dating is introduced in the sidebar on p.135. Teachers might choose to allow it to pass by the way, to discuss with students, or to consider alternative theories they want to share.

● IMPORTANT DATES
Francis Aston (1877-1945) invented a mass spectrograph to identify the isotopes of various elements (1919).
Any dates from the timeline on p.130

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Term 3

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 83**
Matter as Particle and Wave

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.16-17 p.136-149 "Niels Bohr" - "of your view."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- What is quantum mechanics?
- Discuss Bohr and Einstein's friendship. Was it so great because of or despite their difference of opinion? Why?
- Discuss the development of the wave-particle idea.

★ TEACHER NOTE
The end of Ch.16 is missing from the Kindle edition. It should include: "United States, as they will soon learn."

• IMPORTANT DATES
Arthur Compton (1892-1962) demonstrated the particle nature of light (1923).
Louis-Victor de Broglie (1892-1987) showed mathematically that matter has wave properties (1924).
G.P. Thomson (1892-1975) demonstrated the wave nature of electrons (1927).

WEEK 4 ⬚ **30m General Science: Grade 8 - Lesson 84**
Uncertainty Principle

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.18 p.150-157 "The more Niels" - "mentor and poet."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Why is the chapter titled so?
- What statistics are necessary? Why does Einstein dislike this?

★ TEACHER NOTE
The end of Ch.18 is missing from the Kindle edition. It should include: "practical formulas that lead to technological marvels. The founders of quantum mechanics became heroes in an atomic and technological age. Niels Bohr is their mentor and poet."

• IMPORTANT DATES
Werner Heisenberg (1901-1976) introduced the Uncertainty Principle (1925).

WEEK 4 ⬚ **45m Labs: Grade 8 - Lesson 28**
Modeling Atoms

Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of Modeling Atoms, as directed in the Lab Book.

Suggested day 2: Students work on their model.

WEEK 5 ⬚ **30m Natural History: Grade 8 - Lesson 29**
The 'Cure'

Materials: Invincible Microbe

PREP: Read Teacher Tip

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.8 p.91-96 "In 1912," - "in his arms."

★ TEACHER TIP
Are your learners demonstrating more familiarity/interest in the microscopic world? More interest/ease in discussing the relationship between infectious disease and society? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a science museum with a microbiology exhibit.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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Term 3

WEEK 5

⬚ **30m General Science: Grade 8 - Lesson 85**

Quantum Mechanics

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.19-20 p.158-172 "Erwin Schrödinger" - "goes to war."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain the idea of Schrödinger's cat and what he meant to explain by it.
- How did Dirac fuse wave mechanics and particle mechanics?
- Tell what you know about atoms so far.

• IMPORTANT DATES
Erwin Schrödinger (1887-1961) formulated his wave equation to describe electrons (1926).
Schrödinger describes duality as his famous cat (1935).
Paul Dirac (1902-1984), John Cockcroft, and Ernest Walton completed their linear particle accelerator (1932),

WEEK 5

⬚ **30m General Science: Grade 8 - Lesson 86**

Physics and Chemistry

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.20-21 p.172-189 "After the war" - "one way or another."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain why scientists want to accelerate particles and how Einstein's equation is proven.
- Who was Linus Pauling?

• IMPORTANT DATES
Linus Pauling (1901-1994), Francis Crick, and James Watson discovered the chemical structure of DNA (1953).
Any dates from the timeline on p.185

WEEK 5

⬚ **30m General Science: Grade 8 - Lesson 87**

Energy and Mass

Materials: Einstein Adds a New Dimension

➔ RECAP
Briefly recall what you read last time.

➔ READ, NARRATE, & DISCUSS
Ch.22-23 p.190-203 "Scientific thoughts" - "World War II begins."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain the implications of Einstein's equation.
- Discuss the collision of science with World War II. This is a discussion that should continue for many lessons!

★ TEACHER NOTE
Legal cases regarding evolution in the classroom and victimized/marginalized groups are mentioned on p.194, 198, respectively. Teachers might choose to allow these to pass by the way or to discuss with students.

• IMPORTANT DATES
Any events noted in Ch.23 (the chapter serves as an expanded timeline)

WEEK 5

⬚ **45m Labs: Grade 8 - Lesson 29**

Modeling Atoms

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

➔ LAB DAY
Complete day 3 of Modeling Atoms, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook and model.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

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Term 3

	Suggested day 3: Students complete their model and the Analysis and Conclusions section.	
WEEK 6	⬚ **30m Natural History: Grade 8 - Lesson 30** *Antibiotics* ▯ Materials: Invincible Microbe PREP: Read Teacher Tip **→ RECAP** **→ READ, NARRATE, & DISCUSS** Ch.9 p.97-103 "In the autumn" - "wouldn't die."	★ TEACHER NOTE **The age of the Earth is mentioned on p.102:** "millions of years." Teachers might choose to allow it to pass by the way, to decide how they would like to discuss, or to consider alternative theories they would like to share. ★ TEACHER TIP Next week is an activity week! **• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 6	⬚ **30m General Science: Grade 8 - Lesson 88** *The Race to Fission* ▯ Materials: Einstein Adds a New Dimension **→ RECAP** Briefly recall what you read last time. **→ READ, NARRATE, & DISCUSS** Ch.24 p.204-219 "It is September" - "nuclear energy." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Explain what you know about how nuclear fission of the uranium nucleus occurs.	★ TEACHER TIP Remember to engage students in their reading through natural discussion. If you need help, use the bulleted prompts. You can also invite them to share their notebook entries. **• IMPORTANT DATES** Irène and Frédéric Joliot-Curie created the first radioactive isotope artificially (1933). Enrico Fermi split the uranium nucleus (1934). Lise Meitner and Otto Frisch discovered that nuclear fission had occurred in the Hahn lab (1938).
WEEK 6	⬚ **30m General Science: Grade 8 - Lesson 89** *The Bomb* ▯ Materials: Einstein Adds a New Dimension PREP: Read Teacher Tip **→ RECAP** Briefly recall what you read last time. **→ READ, NARRATE, & DISCUSS** Ch.25 p.220-235 "In 1938" - "else possible." Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too. • Describe the scientific race that is beginning.	★ TEACHER TIP Are your learner(s) demonstrating more familiarity/interest in energy or light? More interest/ease in discussing the process of science or the role that curiosity and uncertainty play? If not, consider observing and discussing together more often, exploring extra helpings, or taking a field trip to a planetarium. Reach out through Contact Us if you need help. **• IMPORTANT DATES** Einstein wrote his letter to FDR (1939). Glenn Seaborg discovered plutonium (1940). Self-sustaining nuclear chain reaction (1942)

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Science: Grade 8

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Term 3

WEEK 6

⬚ **30m General Science: Grade 8 - Lesson 90**
The Manhattan Project

▯ Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.26 p.236-253 "In 1943" - "lost its innocence."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Tell the story of the Manhattan Project from the perspective of the scientists.

★ TEACHER NOTE
The end of Ch.26 is missing from the Kindle edition. It should include: "on making the world better, well, science had lost its innocence."

• IMPORTANT DATES
Hiroshima and Nagasaki were bombed (1945).

WEEK 6

⬚ **45m Labs: Grade 8 - Lesson 30**
Measuring the Speed of Light Activity

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 1 of the Measuring the Speed of Light activity as directed in the Lab Book.

Suggested day 1: Students complete the Introduction. They may begin the Procedure, as time permits.

★ TEACHER TIP
Students determine the speed of light empirically using a microwave.

WEEK 7

⬚ **30m Natural History: Grade 8 - Lesson 31**
Antibiotic Activity

▯ Materials: three agar plates, permanent marker, antibiotic discs, small forceps/tweezers, gloves, cotton swabs, rubbing alcohol or hot soapy water

→ INTRO
You just read about the discovery of an antibiotic for use against TB. Let's repeat the same activity from a few weeks ago, but use antibiotic discs this time. How do you think they will compare to the other antimicrobial agents? Do you think all three antibiotics will work the same compared to each other?

→ ACTIVITY
• Turn all of the agar plates upside down and use a marker to divide them into quadrants. Write "Control" in one quadrant of each plate and label each of the remaining quadrants for each antibiotic to be tested. Plate 1 will have the control, and antibiotics A, B, and C in the four quadrants. Plates 2 and 3 will repeat the same sample set. When the plates are all labeled, turn them back over.

• With gloved hands, use a sterile cotton swab to rub the inside of your mouth for 10 seconds. Lift the lid of one plate just enough to evenly and gently rub the cotton swab over the entire surface. Then, recover the plate with the lid. Repeat for all three plates.

• Use rubbing alcohol to sterilize the forceps. Then pick up a single antibiotic disk with the forceps and apply the first of your antimicrobial agents. Remove the lid of one plate and place the disk into the center of the appropriate quadrant. Press gently with the forceps, but do not damage the agar growth medium. Repeat for plates 2 and 3.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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- Clean the forceps with rubbing alcohol and then repeat the last step for the 2 remaining antimicrobial agents and the control. The control will be a plain disc.

- When you are finished, turn all the plates back upside down and incubate in a warm (but not hot) location. You may wish to tape them closed. If you have time to observe them during the week, then do. If not, then they can wait until next week.

WEEK 7

30m General Science: Grade 8 - Lesson 91

Quantum Electrodynamics

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.27-28 p.254-165 "The theoretical" - "they never have."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- How does science proceed after the war?
- Recall Galileo's relativity and explain Einstein's ideas.
- What is the big idea about QED? How does it relate to previous ideas? How does it improve our model of the universe?

WEEK 7

30m General Science: Grade 8 - Lesson 92

Relativity

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.29 p.266-275 "When Isaac" - "you inhabit."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain spacetime.
- Why were previous models of the universe wrong? What key idea was necessary?
- How can stretching the mind and engaging the imagination be helpful in our pursuit of "I AM"?

WEEK 7

30m General Science: Grade 8 - Lesson 93

Spacetime

Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.30-31 p.276-289 "Einstein asks" - "he be thinking?"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Discuss how different realities can exist for different observers,

★ TEACHER TIP
Don't worry if the ideas in the text don't quite make sense - they are challenging! These are lifetime ideas for most of us. Engaging with them to see the big picture is what's important.

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according to relativity.
• Discuss time from our perspective versus time from the perspective of a beam of light.
• Consider that Einstein's thought was steeped in Jewish theology. Discuss any spiritual implications that come to mind.

WEEK 7

45m Labs: Grade 8 - Lesson 31
Measuring the Speed of Light Activity

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 2 of the Measuring the Speed of Light activity as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 2: Students complete the Procedure, as well as the Analysis and Conclusions section.

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 8

30m Natural History: Grade 8 - Lesson 32
Antibiotics

Materials: plates from last week

→ INTRO
Today, you will examine your plates! How well do you think your antimicrobial agents worked?

→ OBSERVE, NARRATE, & DISCUSS
• Retrieve your 3 plates and examine them. Draw or diagram them in your science notebook. Did your antibiotics inhibit growth better than the control? How did they compare to each other? Why do we have different kinds of antibiotics? Can you compare these results with those obtained a few weeks ago on your antimicrobial agents? Why or why not?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 8

30m General Science: Grade 8 - Lesson 94
Speed of Light

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.32-33 p.290-301 "Albert Einstein" - "under your hat."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Continue discussing Einstein's challenging idea!

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

WEEK 8

30m General Science: Grade 8 - Lesson 95
Relative Gravity

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.34 p.302-311 "Gravity. The very" - "with a red flower."

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Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain Einstein's thought experiment regarding gravity.
● What does gravity have to do with relativity?

WEEK 8

30m General Science: Grade 8 - Lesson 96
Spacetime and the Universe

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.35-36 p.312-325 "By 1915" - "surprises are coming."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Explain gravity in terms of spacetime.
● Explain how a solar eclipse proved Einstein's idea.
● Einstein was not perfect. Discuss his acceptance of this fact based on what you have read.

→ SUPPLEMENTAL
∞ Video Link: General Relativity Demo (3:20)

★ TEACHER NOTE
The end of Einstein's first marriage and his relationship with his second wife are introduced on p.317. Ideas concerning the temporal and spatial finiteness of the universe are discussed on p.322-325. Teachers might choose to allow these to pass by the way or to discuss with students.

● IMPORTANT DATES
General relativity was confirmed during the solar eclipse (1919).

WEEK 8

45m Labs: Grade 8 - Lesson 32
Interchanging Mass and Energy Practice

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete Interchanging Mass and Energy Practice, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

If you need some help AFTER you have tried on your own, check out the video link:
∞ Video Link: Lab 4 Einstein (14:21)

★ TEACHER TIP
Students consider the meaning of E=mc^2 by using the equation. Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

WEEK 9

30m Natural History: Grade 8 - Lesson 33
Supergerms

Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.10 p.104-111 "In the records" - "again and again."

● Consider what you learned about natural selection back in Ch.1. Then explain the drug resistance and its solution noted on p.105-106: "The side effects" - "permanent cure."

● OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

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WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 97**
An Expanding Universe

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.37-38 p.326-339 "In 1917" - "thinks it is."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Explain what is meant by red-shifted light.
- Explain why scientists hypothesized that the universe is expanding.
- Discuss Lemaître's ideas.

★ TEACHER NOTE
Ideas concerning the expansion of the universe and the introduction of the Big Bang theory are discussed on p.326-339. Teachers might choose to allow this to pass by the way, to discuss with students, or to consider alternative theories they would like to share.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

• IMPORTANT DATES
Edwin Hubble (1889-1953) discovered other galaxies (1925) and observed the expansion of the universe (1929).

WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 98**
The Fates of Stars

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.39-40 p.340-355 "His real name" - "is at war."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell about Chandra.
- Tell what you know about stars.

• IMPORTANT DATES
Subrahmanyan Chandrasekhar (1910-1995) discovered that certain stars do not become white dwarfs.
Any dates from the timeline on p.352

WEEK 9

⬚ **30m General Science: Grade 8 - Lesson 99**
Black Holes

Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.41 p.356-369 "When supermassive" - "gets his magazine."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

- Tell what you know about black holes.

★ TEACHER NOTE
The Big Bang is considered in the quote on p.359; a bet with a payoff of a "subscription to a racy magazine" is mentioned on p.365; the age of the Earth and the Big Bang are mentioned in the sidebar on p.368. Teachers might choose to allow these to pass by the way or to discuss with students.

• IMPORTANT DATES
Astronauts Armstrong, Aldrin, and Collins landed on the moon (1969).

WEEK 9

⬚ **45m Labs: Grade 8 - Lesson 33**
What Hubble Saw

Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY

★ TEACHER TIP
Students consider the meaning of ‘red shift’ using Hubble’s calculations. Ideas concerning the expansion of the universe are discussed. Teachers might choose to allow this to pass by the way, to discuss with students, or

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Term 3

Complete day 1 of What Hubble Saw, as directed in the Lab Book.

Suggested day 1: Students complete the Introduction today.

to consider alternative theories they would like to share.

WEEK 10 ⬚ **30m Natural History: Grade 8 - Lesson 34**

Supergerms

▯ Materials: Invincible Microbe

→ RECAP

→ READ, NARRATE, & DISCUSS

Ch.11 p.112-117 "There was" - "twenty other people."

• Difficult situations, like the one in this book, can make it difficult for people to treat each other with compassion. Why is that? Could the people in this story have done better? If so, how?

• OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 100**

Gravity Waves?

▯ Materials: Einstein Adds a New Dimension

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.42 p.370-381 "Deep in a pine" - "Stay tuned."

Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Describe the 4 interactions between matter: strong, electromagnetism, weak, and gravity.

★ TEACHER NOTE

The Big Bang and the age of the universe are further considered on p.372-373, 377. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 101**

More Unseen in Space and Time

▯ Materials: Einstein Adds a New Dimension

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.43 p.382-391 "It is 1948" - "with such precision."

Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• What do particles and waves have to do with the "big bang"?

★ TEACHER NOTE

The Big Bang and the age of the universe are further considered on p.382-391. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 10 ⬚ **30m General Science: Grade 8 - Lesson 102**

Inflation and the Theory of Everything

▯ Materials: Einstein Adds a New Dimension

→ RECAP

Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS

Ch.44 p.392-405 "A tiny acorn" - "cosmological system."

Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

★ TEACHER NOTE

The Big Bang and the age of the universe are further considered on p.392-405 and in the optional reading on p.408. Teachers might choose to allow this to pass by the way or to discuss with students.

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Science: Grade 8

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• What is "inflation" to an astrophysicist?

→ NOTE
Read Ch.45 p.406-413 for interest, if desired.

WEEK 10 ⬚ **45m Labs: Grade 8 - Lesson 34**
What Hubble Saw

> Materials: Grade 8 Lab Book and materials listed within

→ LAB DAY
Complete day 2 of What Hubble Saw, as directed in the Lab Book.

Suggested day 2: Students work on the Procedure today.

WEEK 11 ⬚ **30m Natural History: Grade 8 - Lesson 35**
Supergerms

> Materials: Invincible Microbe

PREP: Read Teacher Tip

→ RECAP

→ READ, NARRATE, & DISCUSS
Ch.11 p.117-122 "Surprisingly," - "more comfortably."

• Discuss the need for responsibility in science. Give examples from this or other books that you have read.

★ TEACHER NOTE
The age of the Earth is mentioned on p.102: "millions of years." Teachers might choose to allow it to pass by the way, to decide how they would like to discuss, or to consider alternative theories they would like to share.

★ TEACHER TIP
When students take exams next week, evaluate what they recall, but more importantly, whether microbiology or the relationship between infectious disease and society is more familiar or has possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc?

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 103**
Difficult Predictions

> Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.46 p.414-427 "Brian Schmidt" - "be the limit."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

• Discuss the problem with the theory of expansion.

★ TEACHER NOTE
The Big Bang and the age of the universe are further considered on p.415-425. Teachers might choose to allow this to pass by the way or to discuss with students.

• NATURE NOTEBOOK
Prompt for General Science in Outdoor Work Quick Link.

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WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 104**
Difficult Predictions

▯ Materials: Einstein Adds a New Dimension

PREP: Read Teacher Tip

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.47 p.428-437 "Clocks were" - "it's so long!"
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● Discuss how we use metaphors to help us think about ideas.

★ TEACHER TIP
When student(s) take exams next week, evaluate what they recall, but more importantly whether energy, light, the process of science, or the habits we need to practice as scientists are more familiar or have possibly even become a particular friend! If no relational growth is observed, consider why: do they need more hands-on time, more reading support, less business, etc? Reach out through Contact Us if you need help or have any concerns about readiness for the next stage in relationship building.

WEEK 11 ⬚ **30m General Science: Grade 8 - Lesson 105**
Difficult Predictions

▯ Materials: Einstein Adds a New Dimension

→ RECAP
Briefly recall what you read last time.

→ READ, NARRATE, & DISCUSS
Ch.48-49 p.438-454 "Do you believe" - "yet to discover."
Note ideas that stand out to you in words, pictures, or diagrams. Record any questions or connections you make along the way, too.

● "The classical scientists believed that understanding nature would bring the power of exact prediction." Discuss.
● Discuss why science is about what we don't know rather than what we do.
● Why is the everyday still complex and difficult to predict, despite all that science has learned?
● Consider Einstein's question printed on p.449: "Did God have any choice in the creation of the world?" If the universe has to be the way that it is for "deep mathematical reasons," does that prohibit God's choice and creativity? Or does that rather tell us something about God?
● Do some research to find out how ideas have changed or developed around this question of life on other planets.

★ TEACHER NOTE
The age of the universe is mentioned on p.448. Teachers might choose to allow this to pass by the way or to discuss with students.

WEEK 11 ⬚ **45m Labs: Grade 8 - Lesson 35**
What Hubble Saw

▯ Materials: Grade 8 Lab Book and materials listed within

PREP: Read Teacher Tip

→ LAB DAY
Complete day 3 of What Hubble Saw, as directed in the Lab Book. Be sure to narrate the lab to your teacher by showing them your lab notebook.

Suggested day 3: Students complete the Procedure and Analysis and Conclusions.

Stretching your imagination is much more important than getting the right answers, but if you need some help, you can check out the video link:
∞ Video Link: Lab 5 Einstein (7:53)

★ TEACHER TIP
Be sure to have students share their notebook, during which they should be able to discuss the science and explain what they did.

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Term 3

WEEK 12	⬚ **30m Natural History: Grade 8 - Lesson 36** *Exam Week* **→ EXAMS** Answer question(s) related to course.	**• OUTDOOR WORK** Weekly walks and daily outdoor time are vital! Resources in Quick Links.
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 106** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Einstein Adds a New Dimension	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 107** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Einstein Adds a New Dimension	
WEEK 12	⬚ **30m General Science: Grade 8 - Lesson 108** *Exam Week* **→ EXAMS** Answer question(s) related to the course. Questions will come from: Einstein Adds a New Dimension	
WEEK 12	⬚ **45m Labs: Grade 8 - Lesson 36** **→ EXAMS** Answer question(s) related to course. Questions will come from: Grade 8 Lab Book	

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Examination

Term 1

General Science: Grade 8

· Newton at the Center: Explain how the study of motion helped scientists, like Brahe, Galileo, and Kepler, learn new things about planetary movement.

· Newton at the Center: Explain both perspectives of the conflict between Galileo and the Church.

· Newton at the Center: Explain Newton's 3 Laws of motion using words, drawings, and/or diagrams.

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 8

· Invincible Microbe: Tell what you know about bacteria using any combination of words, drawings, or diagrams.

Term 2

General Science: Grade 8

· Newton at the Center: Newton, Boyle, Cavendish, Lavoisier, and many others made important discoveries about matter. Explain how their work helped transition alchemy into a true science, called chemistry.

· Newton at the Center: Explain 3 ways that moving particles can affect other matter.

· Newton at the Center: Discuss the relationship between electricity and magnetism.

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 8

· Invincible Microbe: Explain some of the challenges in combating TB in the early days of the epidemic.

Term 3

General Science: Grade 8

· Einstein Adds a New Dimension: Tell what you know about the nature of light.

· Einstein Adds a New Dimension: Explain what is meant by E = mc^2 and give examples as to how this is relevant in the world.

· Einstein Adds a New Dimension: Describe the process of science, using examples from the text, as needed.

· Explain how you did one laboratory investigation this term and what you learned from it.

Natural History: Grade 8

· Invincible Microbe: Describe how drug resistance occurs. OR Tell what scientists learned from the TB epidemic.

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